26 Summer Research Final Report:

Scattering Amplitudes, Asymptotic Analysis, and the Emergence of Viscosity Cross Section in Light-Mediator Self-Interacting Dark Matter

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Abstract

This research report presents a systematic evaluation of sub-GeV Dirac fermion dark matter (χ) interacting via a light vector mediator (A'), focusing specifically on testing and establishing the precise regime of validity for Equation (B2) in Knapen, Lin, & Zurek (arXiv:1709.07882, Appendix B). Using `Mathematica` and `FeynCalc`, we calculated the squared cross sections $|M|^2$ for seven leading-order (LO) scattering processes, and verified Wick's theorem contraction signs and neutrino spin degrees of freedom. Through systematic asymptotic analysis in the dimensionless ratio parameter $R \equiv m_\chi v / M_A$, we diagnosed the unphysical divergence of the standard momentum transfer cross section σ_T (weighted by $1-\cos\theta$) in identical dark particles (and also anti-dark particles) scattering under the light-mediator limit ($R \gg 1$), which induces a divergent discrepancy (slope -1 in LogLog space) against Eq. (B2) (slope 0). Grounded in the kinetic theory of gases (Chapman & Cowling, 1939) and the moments of the Boltzmann collision integral, we demonstrate that forward-backward symmetry in identical particle collisions causes the vector momentum moment $m\vec{v}$ to degenerate, naturally transitioning the transport physics to the traceless symmetric second-order velocity tensor moment $\psi_{ij} = m(v_i v_j - \frac{1}{3}v^2\delta_{ij})$, since from the physical picture of halos, we want a fluid-dynamic style of description and show the momentum dissipation in each direction. This yields the viscosity weighting $\sin^2\theta$, removing the $R \gg 1$ divergence and yielding an analytical ratio $\alpha_{B2}/\alpha_{\sigma_V} = 1/\sqrt{8} \approx 0.353553$, thereby completely explaining the physical origin of the factor 8 in Eq. (B2).

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1 Previous Work and Motivation: Testing Formula (B2) in Knapen et al. (1709.07882)

In standard cold dark matter cosmology, collisionless dark matter successfully accounts for large-scale structure formation. However, at sub-galactic scales ($\lesssim 10$ kpc), N-body simulations predict sharp core density profiles (Core-Cusp problem) and an overabundance of massive subhalos (Too-Big-To-Fail problem), which conflict with observed dwarf galaxy velocity dispersions. Self-Interacting Dark Matter (SIDM) resolves these small-scale structure anomalies through elastic self-scattering ($\sigma/m_\chi \sim 0.1 - 10$ cm^2/g), where frequent collisions in dense halo cores thermalize the central velocity distribution.

Achieving such large scattering cross sections ($\sigma/m_\chi \sim 10^{-24}$ cm^2/GeV) for sub-GeV dark matter without over-annihilating in the early universe requires introducing a light dark force mediator (e.g., dark photon A' with mass $M_A \ll m_\chi$).

The primary physical objective of this research project is to **test, verify, and establish the exact boundary and regime of validity** for the formula in Appendix B of Knapen, Lin,

& Zurek (arXiv:1709.07882), specifically Equation (B2):

$$\sigma_T^{\text{Born}} \approx \frac{8\pi\alpha_X^2}{m_\chi^2 v^4}\left(\ln(1+R^2) - \frac{R^2}{1+R^2}\right), \quad \text{where } R \equiv \frac{m_\chi v}{M_A}. \tag{1}$$

Our study addresses two core questions regarding Eq. (1):

1. Does Eq. (1) directly originate from the standard momentum transfer cross section σ_T (weighted by $1-\cos\theta$), or does it rely on alternative angular weighting assumptions?
2. What is the exact physical and algebraic origin of the numerical factor 8, and under what kinematic limits ($R \ll 1$ vs. $R \gg 1$) does this approximation remain valid for identical particle self-scattering?

2 Research Trajectory, Calculations, and Technical Discoveries

2.1 Matrix Element Calculations and Computational Debugging (Research Notes 1 & 2)

We consider a Dirac fermion dark matter candidate χ coupled to a vector mediator A'_μ with coupling constant g_χ, described by the interaction Lagrangian:

$$\mathcal{L}_{\text{int}} = g_\chi \bar{\chi}\gamma^\mu \chi A'_\mu + \sum_f g_f \bar{f}\gamma^\mu f A'_\mu. \tag{2}$$

Using `Mathematica` with the `FeynCalc` package, we systematically evaluated the squared Feynman matrix elements $|M|^2$ for seven leading-order (LO) tree-level scattering processes (averaging initial states, and summing final states):

1. $\chi\bar{\chi} \to \chi\bar{\chi}$ (Dark matter-antidark matter scattering)
2. $\chi\chi \to \chi\chi$ (Identical fermion dark matter self-scattering)
3. $\bar{\chi}\bar{\chi} \to \bar{\chi}\bar{\chi}$ (Antidark matter self-scattering)
4. $\mu^-\mu^+ \to \chi\bar{\chi}$ (Muon annihilation to dark matter)
5. $\tau^-\tau^+ \to \chi\bar{\chi}$ (Tau annihilation to dark matter)
6. $\nu_\mu\bar{\nu}_\mu \to \chi\bar{\chi}$ (Muon neutrino annihilation)
7. $\nu_\tau\bar{\nu}_\tau \to \chi\bar{\chi}$ (Tau neutrino annihilation)

During the symbolic implementation and verification (checking with the well-known form of classical processes under certain limits.), several critical physical and computational issues were identified and resolved:

- **s-Channel and its influence in Process 1:** We identified that because χ and $\bar{\chi}$ are distinguishable particles and antiparticles, an s-channel annihilation exchange diagram M_{1s} must be included alongside the t-channel diagram M_{1t}, along with their interference terms. This correction was confirmed to contribute significantly in both non-relativistic and relativistic regimes. And this lead to the Fermion Exchange Sign problem. When validating the massless limit against standard Bhabha scattering ($e^+e^- \to e^+e^-$), an overall relative sign discrepancy appeared ($M_{1s} - M_{1t}$ vs. $M_{1s} + M_{1t}$). By performing explicit operator contraction sequencing according to Wick's theorem, we demonstrated that fermion interchange between s- and t-channel contractions produces a relative minus sign, verifying the physical correctness of the derivation.

- **Diagnosis and Resolution of `SetMandelstam` Cross-Contamination:** When automating the computation across Processes 1–7, an anomaly appeared in the crossing symmetry checks for Processes 4–7. Tracing the internal code logic revealed that `FeynCalc`'s `SetMandelstam` function creates global scalar product substitution rules in `$Global`. When switching particle mass assignments (e.g., from m_χ to m_μ), stale rules persisted, causing hidden variable contamination. We initially implemented block separation to isolate workspace variables, and subsequently adopted explicit substitution rules via `GetMandelstamRules[...]`, permanently resolving the bug.
- **Neutrino Spin-Averaging and Code Restructuring:** In initial `FeynCalc` implementations for Processes 6 and 7 ($\nu\bar{\nu} \to \chi\bar{\chi}$), an omitted 1/2 in $P_L = \frac{1-\gamma^5}{2}$ was manually compensated during spin-averaging. Recognizing the unreliability of this "too clever" shortcut, I rewrote the computation from scratch using explicit P_L rules. Furthermore, a discussion with Prof. Trickle resolved an apparent discrepancy in the spin-average factor (1/4 vs. 1/1): dividing by 4 applies to the inverse process ($\chi\bar{\chi} \to \nu\bar{\nu}$, arXiv:2603.03444), whereas our forward process ($\nu\bar{\nu} \to \chi\bar{\chi}$) has only 1 physical helicity state per left-handed neutrino ($1 \times 1 = 1$).

2.2 Asymptotic Analysis and Failure of the Transfer Cross Section σ_T (Research Notes 2, 3, 4, 4.5, & 5)

To evaluate the limits of Eq. (1), we constructed an exact non-relativistic "Mother Formula" in `Mathematica` as a function of the dimensionless mass-velocity ratio $R \equiv \frac{m_\chi v}{M_A}$.

We then performed systematic asymptotic series expansions using `Series[..., {R, 0, 4}]` for the heavy-mediator limit ($R \ll 1$) and `Series[..., {R, Infinity, 0}]` for the light-mediator limit ($R \gg 1$).

2.2.1 The $R \ll 1$ Limit (Heavy Mediator)

For $R \ll 1$, the analytical expansion yields a constant, velocity-independent cross section corresponding to standard contact Born scattering:

$$\sigma_T^{(R\ll 1)} \propto \frac{\alpha_X^2 m_\chi^2}{M_A^4}, \tag{3}$$

which matches both Eq. (1) and established literature bounds.

2.2.2 The $R \gg 1$ Limit (Light Mediator) and R^2 Divergence

For identical fermion scattering (Process 2: $\chi\chi \to \chi\chi$), u-channel exchange introduces a Coulomb scattering pole in the backward direction ($\theta \to \pi$, i.e., $t \leftrightarrow u$).

The standard momentum transfer cross section is defined with a forward angular regulator:

$$\sigma_T \equiv \int \frac{d\sigma}{d\Omega}(1 - \cos\theta) d\Omega. \tag{4}$$

While $(1 - \cos\theta) \to 0$ as $\theta \to 0$ successfully regularizes the forward t-channel pole, at backward angles $\theta \to \pi$, $(1 - \cos\pi) = 2$. Consequently, the factor $(1 - \cos\theta)$ fails to suppress the u-channel backward divergence.

Carrying out the $R \gg 1$ asymptotic expansion for σ_T yields:

$$\sigma_{T2}^{(R\gg 1)} \propto \frac{\pi \alpha_X^2}{m_\chi^2 v^4} \left[4R^2 - 8\ln R + \mathcal{O}(1)\right]. \tag{5}$$

Because of the unphysical R^2 term in Eq. (5), plotting the coupling constant α_X against R in LogLog space produces an asymptotic slope of -1 ($\alpha_X \propto R^{-1}$). In contrast, Eq. (1) yields a flat slope of 0 ($\alpha_X \propto$ const).

Taking the ratio between the derived coupling constant α_{σ_T} and α_{B2} gives:

$$\lim_{R\to\infty} \frac{\alpha_{\sigma_T}}{\alpha_{B2}} \to \infty. \tag{6}$$

This diagnostic asymptotic analysis conclusively proved that Eq. (B2) cannot be derived from the standard transfer cross section σ_T.

2.2.3 Order of Limits and Plotting Conventions

During this asymptotic analysis, two methodological insights emerged:

- **Order of Limits and Infrared Structure:** To consistently probe the non-relativistic light-mediator regime ($v \ll 1$ with $R \equiv m_\chi v/M_A \gg 1$), substituting $M_A = m_\chi v/R$ prior to asymptotic expansion is essential; taking $v \to 0$ with fixed M_A would erroneously force $R \to 0$ into the heavy-mediator limit. Furthermore, retaining the sub-leading $\ln R$ term is physically vital: while switching to σ_V eliminates the unphysical R^2 power divergence of σ_T, the residual $\ln R$ term represents the true Coulomb logarithm. For the result when switching to σ_V, see section 2.3.4.
- **Shaded Region Plotting Conventions:** In initial visualizations, shaded areas were drawn to denote "allowed parameter space". Following discussions with Prof. Trickle, the convention was updated to conform with high-energy physics literature standards: **shaded regions represent excluded parameter space**, while white regions represent viable model space.

2.3 Emergence of Viscosity Cross Section σ_V and Origin of Factor 8 (Research Notes 4.5, 5, 6, & 7)

To resolve the backward divergence of identical particle scattering and establish the physical foundation of Eq. (1), we investigated the angular weighting function from three gradually perspectives.

2.3.1 Algebraic Symmetrical Regularization

Since identical particle scattering is symmetric under forward-backward interchange ($\theta \leftrightarrow \pi-\theta$), a regularizing weighting function must simultaneously vanish at $\theta \to 0$ and $\theta \to \pi$. Multiplying the forward regulator $(1-\cos\theta)$ by the backward regulator $(1+\cos\theta)$ constructs the symmetric product:

$$F(\theta) = (1-\cos\theta)(1+\cos\theta) = 1-\cos^2\theta = \sin^2\theta. \tag{7}$$

This algebraically solves the observation redundancy, and naturally defines the **Viscosity Cross Section σ_V**:

$$\sigma_V \equiv \int \frac{d\sigma}{d\Omega} \sin^2\theta \, d\Omega. \tag{8}$$

Note that $1-\cos\theta$ & $1-\cos^2\theta$ looks like Legendre Polynomials, which implies the angular behaviors. This will be used in the derivation in section 2.3.3.

2.3.2 Kinetic Dissipation and Geometry of Traceless Velocity Tensors

Following Chapman & Cowling (1939) (*The Mathematical Theory of Nonuniform Gases*), the viscous dissipation rate in microscopic gas collisions is governed by changes in the traceless second-order velocity tensor $Q_{ij} \equiv v_i v_j - \frac{1}{3}v^2\delta_{ij}$.

For a binary collision with relative velocity g, calculating the norm-squared change in Q_{ij} integrated over azimuthal angle ϕ gives:

$$\int_0^{2\pi} d\phi \, \|\Delta Q_{ij}\|^2 = 2g^4 \sin^2\theta, \tag{9}$$

demonstrating that transverse momentum shear and viscous dissipation are geometrically dictated by the $\sin^2\theta$ weighting factor.

2.3.3 Boltzmann Collision Integral and Moment Equations

From the Boltzmann transport equation $\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_r f = C[f]$, taking velocity-space moments $\int \psi(\vec{v}) C[f] d^3v$ generates macroscopic transport relations:

1. **Vector Moment $\psi(\vec{v}) = m\vec{v}$ (Longitudinal Momentum Transfer):** Governs momentum loss along the initial relative velocity vector, generating the $(1 - \cos\theta)$ weighting of σ_T. For identical particles ($\chi\chi \to \chi\chi$), forward and backward collisions are quantum mechanically indistinguishable, causing the first-order vector momentum flux to degenerate.

2. **Traceless Tensor Moment $\psi_{ij}(\vec{v}) = m(v_i v_j - \frac{1}{3}v^2\delta_{ij})$ (Viscous Stress / Transverse Shear):** Represents the $l = 2$ spherical harmonic mode of the collision operator. In identical particle scattering, transport physics **naturally emerges** through this second-order tensor moment, which rigorously isolates transverse shear and produces the $\sin^2\theta$ angular weighting of σ_V.

2.3.4 Analytical Validation and Resolution of Factor 8

Evaluating Process 2 ($\chi\chi \to \chi\chi$) using the viscosity cross section σ_V with the identical particle phase space factor 1/2 in the heavy-mediator limit ($R \ll 1$) gives the expected result (using transfer cross section also gives correct result in this limit),
and in the light-mediator limit ($R \gg 1$) yields:

$$\sigma_V^{(R\gg1)} = \frac{2\pi\alpha_X^2}{m_\chi^2 v^4}\left[\ln(1+R^2) - \frac{R^2}{1+R^2}\right]. \tag{10}$$

Eq. (10) completely eliminates the unphysical R^2 term, recovering a flat asymptotic slope of 0 in LogLog parameter space.

Comparing Eq. (10) with Knapen et al. Eq. (1):

$$\sigma_T^{\text{Born (B2)}} = \frac{8\pi\alpha_X^2}{m_\chi^2 v^4}\left[\ln(1+R^2) - \frac{R^2}{1+R^2}\right] = 4 \times \sigma_V^{(R\gg1)}. \tag{11}$$

Equating cross sections to extract the coupling constant relation gives:

$$\alpha_{B2} = \frac{1}{\sqrt{4}}\alpha_{\text{uncorrected}} \quad \Longrightarrow \quad \frac{\alpha_{B2}}{\alpha_{\sigma_V}} = \frac{1}{\sqrt{8}} \approx 0.353553. \tag{12}$$

This exact analytical ratio proves that Equation (B2) in Knapen et al. (1709.07882) is **precisely grounded in the viscosity cross section σ_V for identical particle scattering**, where the factor 8 represents the exact combination of the $l = 2$ tensor moment angular integration and the 1/2 identical particle phase space symmetry factor.

Now we had verified the accuracy of equation B2 in the literature. It described the corrected behavior for Processes 1,2,3 as we listed in section 2.1.

2.4 Regime of Validity

The matrix elements, asymptotic limits, and cross sections derived in this report are established within a precise quantum field theory framework defined by the following model assumptions and kinematic regimes:

1. **Lagrangian and Field Content:** We consider Dirac fermion dark matter (χ) coupled to a vector mediator (A'_μ) or a scalar boson (ϕ), governed by the interaction Lagrangians:

$$\mathcal{L}_{\rm int}^{\rm vector} = g_\chi \bar{\chi}\gamma^\mu \chi A'_\mu + \sum_f g_f \bar{f}\gamma^\mu f A'_\mu, \quad \text{or} \quad \mathcal{L}_{\rm int}^{\rm scalar} = y_\chi \bar{\chi}\chi\phi + \sum_f y_f \bar{f} f \phi. \tag{13}$$

2. **Initial-State Averaging and Final-State Summation:** All squared Feynman matrix elements $\overline{|M|^2}$ are evaluated at tree level (Leading Order) for unpolarized scattering by explicitly summing over final-state spin configurations and averaging over initial-state spin/helicity degrees of freedom:

$$\overline{|M|^2} = \frac{1}{g_1 g_2} \sum_{\rm spins} |M|^2, \tag{14}$$

   where $g_i = 2$ for standard unpolarized Dirac fermions, and $g_i = 1$ for left-handed neutrinos possessing a single physical helicity state.

3. **Non-Relativistic Kinematics ($v \ll 1$):** The microphysical cross sections are derived in the center-of-mass frame under the non-relativistic limit ($v \ll 1$), systematically neglecting $\mathcal{O}(v^2)$ relativistic recoil corrections.

4. **Perturbative Born Boundary ($\alpha_X m_\chi / M_A \ll 1$):** The derived matrix elements and formula (B2) hold strictly within the perturbative Born approximation ($\alpha_X m_\chi / M_A \ll 1$). The regime of $\alpha_X m_\chi / M_A \gtrsim 1$, where non-perturbative Sommerfeld enhancements and bound states dominate, lies beyond this tree-level validity boundary.

3 Future Research Directives

Based on the theoretical framework established in this project, future investigations will focus on the following directives:

1. **Extend the regime of validity:** The most natural thing to do next is to generalize the theory outside of the regime of validity we described above. For example, extend to real/complex scalar fermion dark matter, and non-vector mediators (scalar, pseudoscalar, and axial-vector).

2. **Emergence of viscosity cross section and the SIDM limit:** Now the reason of switching from transfer cross section to viscosity cross section is the observation redundancy (degeneracy), but we expect a mathematical description from the methodology starting from the vector moment, applied with identical particles, and get an unacceptable result, then the tensor moment naturally arises. Furthermore, we expect the SIDM limit will naturally emerge from the derivation starting from the moment of Boltzmann Equation.

3. **Higher-Order Boltzmann Tensor Moments:** Since we are deriving viscosity cross section from a higher order moment than that for transfer cross section, it is natural to derive, for example, third-order anisotropic moments ψ_{ijk} to investigate whether non-equilibrium, highly anisotropic dark matter halos require generalized transport cross section sequences $\sigma^{(n)}$.

4. **Direct Numerical Boltzmann Integration vs. N-Body Simulations:** It is hard to solve Boltzmann Equation, so we aimed to solve its moments. But as the technology develops, Solve the full velocity-space Boltzmann equation without moment closure approximations might be possible. After this, use the result to test halo density profile evolution against SIDM N-body simulations.

5. **Thermal Averaging and Non-Perturbative Resonances:** There is one thi g mentioned in the beginning of the research, but haven't done yet. We had 7 processes in the beginning, but only discussed te first 3 of them. The last 4 should be considered for the following process: Compute phase-space thermally averaged cross sections $\langle \sigma v \rangle(T)$ incorporating non-perturbative Sommerfeld enhancement and quantum mechanical resonance peaks in the sub-GeV regime.

Summary Statement: *By computing the exact microscopic tree-level scattering amplitudes for sub-GeV Dirac dark matter with a vector mediator and performing rigorous asymptotic analysis, this work identified the failure of standard transfer cross sections in identical particle light-mediator scattering and demonstrated the natural emergence of the viscosity cross section σ_V from Boltzmann tensor moments, establishing the precise regime of validity and physical origin of factor 8 in Eq. (B2) of arXiv:1709.07882.*